

Abdurrahman Shaffar

QUANTUM RESEARCH ENGINEER · APPLIED RND ENGINEER

Suwon-si, Gyeonggi-do, Republic of Korea

✉ abdurrahman [at] shaffar [dot] com | 🏠 shaffar.com | 📧 awshaff | 📺 awachid | 🎓 Google Scholar

“Toward Quantum-Enhanced Solutions for Industry Challenges”

Summary

A quantum communications researcher focused on stochastic channel modeling, entanglement distribution, and fading analysis for microwave-based quantum systems essential for distributed quantum computing. Studying continuous-variable quantum information approaches that bridge classical communication theory and quantum-statistical performance, with the goal of building infrastructure for tomorrow's quantum networks. A former backend software engineer, bringing practical experience in distributed systems, API development, and scalable architecture to the design of robust quantum communication solutions. Currently, a PhD researcher at CQILAB, Kyung Hee University, Republic of Korea, under the supervision of Professor Shin Hyundong. Previously worked as a backend software engineer, delivering projects across billing systems, KYC, e-commerce, e-procurement, and HRIS domains using PHP, RabbitMQ, Redis, and MinIO.

Highlights

- Published across top-tier venues in robotics (IEEE/RSJ IROS, IEEE ICRA, IEEE RA-L, RSS, IEEE T-RO, IJRR), autonomous driving (IEEE IV, IEEE T-IV), and artificial intelligence (CVPR, ICCV, AAAI, NeurIPS, IEEE T-PAMI).
- Always pursue *clean, useful, reproducible* code to support the community:
 - e.g., Maintainer of TEASER++ [📄 ★ 2,100] and Hydra [📄 ★ 773] & publisher of Patchwork++ [📄 ★ 842] and Patchwork [📄 ★ 555], KISS-Matcher [📄 ★ 643], VGGT-SLAM [📄 ★ 882], ERASOR [📄 ★ 494], and BUFFER-X [📄 ★ 243]
- Academic honors & awards:
 - IEEE ICRA 2025 Outstanding Reviewer Award (Selected as one of only 5 awardees out of 7,641 reviewers)
 - RSS Pioneers 2024 (Out of 202 candidates, only 30 researchers were selected)
 - 2022 IEEE RA-L Best Paper Award (among 1,100 papers, only 5 papers were selected)
- Won multiple SLAM challenges and competitions while pursuing technologies for real-world applications:
 - 1st prize again in the HILTI SLAM Challenge'24 in IEEE ICRA
 - 1st prize in HILTI SLAM Challenge'23 in IEEE ICRA among 63 international teams
 - 2nd cash prize in HILTI SLAM Challenge'22 in IEEE ICRA (in total, 4th place)
 - 1st prize in Hitachi-LG Data Storage LiDAR application competition, Republic of Korea
 - Received CES'23 innovation award through technology transfer on SLAM for mobile robots (collaborated with HILLS Robotics)
- Professional experience:
 - Postdoctoral associate of Laboratory for Information & Decision Systems (LIDS), MIT (advisor: Prof. Luca Carlone)
 - Visiting scholar of Univ. Bonn, Germany (advisor: Prof. Cyrill Stachniss)
 - In 2022, served as a SLAM part outside expert, CTO division of LG Electronics, Republic of Korea
 - Research intern in vision/deep learning team at NAVER LABS, Republic of Korea

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Continuous-variable quantum information and quantum microwave communication
- Stochastic channel modeling for quantum wireless links
- Entanglement distribution and degradation over noisy channels
- Fading analysis in open-air and ground-to-ground microwave systems
- Rydberg atomic receivers for quantum signal detection
- Quantum computing and distributed quantum information systems

Bridging classical communication theory with quantum-statistical system design

Education

Kyung Hee University (KHU)

PH.D. IN ELECTRONICS AND INFORMATION CONVERGENCE ENGINEERING

Yongin, Republic of Korea

Mar. 2022 - Feb. 2028

- Dissertation: “Quantum Microwave Communication”

Universitas Gadjah Mada (UGM)

B.S. IN PHYSICS

Yogyakarta, Republic of Indonesia

Aug, 2015 - Nov. 2019

- Dissertation: “Quantum Mechanics Implementation in Quantum Cryptography”
- Advised by Prof. Pekik Nurwantoro

Work Experience (Including Internship)

Postdoctoral Associate

Massachusetts, the U.S.

SPARK LAB, MIT

Apr. 2024 - Present

- Advised by Prof. Luca Carlone.
- Conduct research on multi-robot/-session autonomous mapping algorithms in complex urban environments.
- Guide and supervise graduate and undergraduate students in their research papers and thesis projects.

SEPTEMBER 18, 2026

ABDURRAHMAN WACHID SHAFFAR · CURRICULUM VITAE

1

Postdoctoral Fellow

URBAN ROBOTICS LAB, KAIST

- Advised by Prof. Hyun Myung.
- Conducted research on localization and mapping techniques using LiDAR and vision data, respectively.
- Developed deep learning models for robust perception.
- Guided and supervised graduate and undergraduate students in their research papers and thesis projects.

Daejeon, Republic of Korea

Mar. 2023 - Mar. 2024

Outside Expert of the CTO Division of LG Electronics Co. Ltd.

LG ELECTRONICS CO. LTD.

- Introduced state-of-the-art SLAM technologies and provided insights into future technological directions for mobile robots.
- Delivered a lecture entitled "Introduction to the latest SLAM technology trends and Semantic SLAM."

Republic of Korea

Mar. 2022 - Feb. 2023

Visiting Scholar

STACHNISSLAB, UNIVERSITY OF BONN

- After defending my dissertation, I visited Prof. Cyrill Stachniss' lab.
- Worked on an improved version of static map building.
- Submitted and had a paper accepted at RSS 2023 titled ERASOR2.

Bonn, Germany

Nov. 2022 - Feb. 2023

Research Intern

NAVER LABS

- Served as a Ph.D. research intern.
- Worked on robust global registration using 3D point cloud data captured by the LiDAR sensor.
- Submitted and accepted a paper to IEEE ICRA 2022.
- Relevant code: <https://github.com/url-kaist/Quatro>.

Gyeonggi-Do, Republic of Korea

Apr. 2022 - Sep. 2022

Research Project Experience

Development of Harmonic-Space Operation Using Seamless Information Sharing Among Large-Scale Heterogeneous UVs

SUPPORTED BY NATIONAL RESEARCH FOUNDATION, REPUBLIC OF KOREA

- Implemented multi-session LiDAR SLAM
- Studied semantic SLAM for heterogeneous robot platforms

Keywords: 3D semantic mapping, Scene graph-based SLAM

SPARK Lab, MIT

Sep. 2024 - Current

Research on Certifiable Perception and Long-Term Metric-Semantic Mapping for Autonomous Vehicles

SUPPORTED BY NATIONAL RESEARCH FOUNDATION, REPUBLIC OF KOREA

- Studied long-term metric-semantic mapping for mobile robots or autonomous vehicles

Keywords: Metric semantic mapping, Semantic SLAM

Postdoctoral Fellowship Program

Sep. 2024 - Sep. 2025

Development of Scene Graph-Based Multi-Robot Multi-Session SLAM

SUPPORTED BY DCIST & DEVCOM

- Implemented scene graph-based SLAM methods based on Hydra
- Ran semantic SLAM and enhanced the performance in our real-world campus scenes

Keywords: 3D semantic mapping, Semantic SLAM

SPARK Lab, MIT

Apr. 2024 - Current

Development of Semantic SLAM for Spatial AI

SUPPORTED BY LG ELECTRONICS

- Implemented deep learning-aided SLAM methods based on Hydra
- Ran semantic SLAM and enhanced the performance in our real-world campus scenes

Keywords: 3D semantic mapping, Semantic SLAM

Urban Robotics Lab., KAIST

Apr. 2023 - Dec. 2023

Development of Artificial Intelligence Robot Autonomous Navigation Technology for Agile Movement in Crowded Space

SUPPORTED BY THE MINISTRY OF TRADE, INDUSTRY & ENERGY (MOTIE), REPUBLIC OF KOREA

- Was in charge of dynamic object detection in order to generate static maps and perform moving-object-robust SLAM in dynamic environments

Keywords: SLAM, 3D LiDAR, Static Map Building, OpenCV, C++, ROS

Urban Robotics Lab., KAIST

Mar. 2020 - June 2023

Study of Semantic SLAM Towards Spatial AI Technologies

SUPPORTED BY LG ELECTRONICS

- Studied various deep learning-aided SLAM methods, such as Kimera and Hydra
- Run semantic SLAM in our real-world data to check the feasibility

Keywords: 3D semantic mapping, Semantic SLAM

Urban Robotics Lab., KAIST

Mar. 2022 - Dec. 2022

Deep Learning-Based Depth Prediction Using a Mono Camera and 2D LiDAR Sensor

SUPPORTED BY HYUNDAI KEFICO

- Was in charge of the development of deep learning-based depth prediction
- Studied various calibration methods (2D LiDAR-to-camera, 3D LiDAR-to-IMU, camera-to-IMU)
- Ran deep learning models on NVIDIA Xavier

Keywords: Real-time deep learning, Depth prediction, Calibration, PyTorch, OpenCV, C++, ROS

Urban Robotics Lab., KAIST

Mar. 2022 - Nov. 2022

A Study on the Visual Place Recognition in Multiple Photos

SUPPORTED BY KAIST INSTITUTE FOR SECURITY CONVERGENCE RESEARCH

- Studied visual place recognition application. Struggled to improve the performance of the SOTA deep learning-based VPR approaches

Keywords: Visual place recognition, Deep learning, Semantic segmentation, Inpainting, Python, PyTorch

Urban Robotics Lab., KAIST

Mar. 2020 - Dec. 2020

Development of Robot Intelligence Technology for Mobility with Learning Capability Toward Robust and Seamless Indoor and Outdoor Autonomous Navigation

SUPPORTED BY THE MINISTRY OF TRADE, INDUSTRY & ENERGY (MOTIE)

- Developed SLAM algorithm applied to mobile robots
- Was in charge of SLAM, static map building in low-dynamic environments, and movable area prediction

Keywords: SLAM, 3D LiDAR, Registration, Static map building, Mobile robots, C++, ROS

Urban Robotics Lab., KAIST

Mar. 2020 - Sep. 2020

Indoor Navigation of Mobile Robots using Deep Learning-based Object Recognition

Urban Robotics Lab., KAIST

SUPPORTED BY SAMSUNG ELECTRONICS CO., LTD.

Jan. 2019 - Sep. 2019

- Developed SLAM and perception algorithms applied to robot cleaners for achieving robust navigation in indoor environments
- Was in charge of depth prediction using a 2D LiDAR sensor and a monocular camera for collision avoidance of mobile robots via deep learning

Keywords: 2D LiDAR, Sensor Fusion, Deep Learning, Mobile Robots, PyTorch, ROS

IITP Artificial Intelligence R&D Grand Challenge: Track 4, Intelligent Control

Urban Robotics Lab., KAIST

SUPPORTED BY IITP, WHICH IS A GOVERNMENT-AFFILIATED ORGANIZATION

Jan. 2019 - Jun. 2019

- Was in charge of the task of a drone passing through windows
- Implemented RGB-D camera-based path planning&following. Participated in applying VIO to estimate the odometry of UAV

Keywords: VIO, Path planning and following, Projective geometry, OpenCV, ROS

Machine Learning-Based Classification of Small Object Captured by Unmanned Aerial Vehicle

Urban Robotics Lab., KAIST

OUTSOURCED BY PIXONEER GEOMATICS AND AGENCY FOR DEFENSE DEVELOPMENT

Jan. 2018 - Dec. 2019

- Developed both SVM-based and Deep Learning-based classification algorithms
- Implemented HOG-LBP for input to SVM and engaged in designing novel Deep Learning architecture.

Keywords: Deep Learning, SVM, HOG-LBP, Classification, Python, PyTorch

Range-Only SLAM in Complex Disaster Situation

Urban Robotics Lab., KAIST

SUPPORTED BY MINISTRY OF TRADE, INDUSTRY, AND ENERGY

Jan. 2018 - Dec. 2018

- Implemented Monte Carlo localization (MCL) using range measurements by ultra-wideband (UWB) sensors for UAV from scratch single-handed
- Struggled to cover Non-line-of-sight (NLOS) issues.
- Keywords: MCL, Beacon-based localization, UWB sensors, NLOS, ROS

INTERNATIONAL JOURNAL

1. **Hyungtae Lim**, Minkyun Seo, Luca Carlone, and Jaesik Park, “Towards Zero-Shot Point Cloud Registration Across Diverse Scales, Scenes, and Sensor Setups,” in *IEEE Trans. Pattern Anal. Mach. Intell. (T-PAMI)*, 2026. Under review.
2. **Hyungtae Lim**, Daebeom Kim, and Hyun Myung, “Multi-Mapcher: Loop Closure Detection-Free Heterogeneous LiDAR Multi-Session SLAM Leveraging Outlier-Robust Registration for Autonomous Vehicles,” *IEEE Trans. Intell. Veh. (T-IV)*, vol. 11, no. 2, pp. 338–351, 2026
3. I Nahrendra, Byeongho Yu, Minho Oh, Dongkyu Lee, Seunghyun Lee, Hyeonwoo Lee, **Hyungtae Lim**, and Hyun Myung, “Obstacle-Aware Quadrupedal Locomotion With Resilient Multi-Modal Reinforcement Learning,” *IEEE Trans. Robot. (T-RO)*, vol. 42, pp. 819–836, 2026.
4. Kevin Christiansen Marsim, Minho Oh, Byeongho Yu, Seungjae Lee, I Made Aswin Nahrendra, **Hyungtae Lim**, and Hyun Myung, “LVI-Q: Robust LiDAR-Visual-Inertial-Kinematic Odometry for Quadruped Robots Using Tightly-Coupled and Efficient Alternating Optimization,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 10, no. 10, pp. 10050–10057, 2025.
5. Daehan Lee, **Hyungtae Lim**, and Soohee Han, “GenZ-ICP: Generalizable and Degeneracy-Robust LiDAR Odometry Using an Adaptive Weighting,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 10, no. 1, pp. 152–159, Nov. 14, 2024.
6. Seungwon Song, **Hyungtae Lim**, Alex Junho Lee, and Hyun Myung, “DynaVINS++: Robust Visual-Inertial State Estimator in Dynamic Environments by Adaptive Truncated Least Squares and Stable State Recovery,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 9, no. 10, pp. 9127–9134, Sep. 6, 2024.
7. DongKi Noh, **Hyungtae Lim**, Gyuho Eoh, Duckyu Choi, Jeong-Sik Choi, Hyunjun Lim, Seung-Min Baek, and Hyun Myung, “CLOi-Mapper: Consistent, Lightweight, Robust, and Incremental Mapper With Embedded Systems for Commercial Robot Services,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 9, no. 9, pp. 7541–7548, Jul. 15, 2024.
8. **Hyungtae Lim**, Minho Oh, Seungjae Lee, Seunguk Ahn, and Hyun Myung, “Similar but Different: A Survey of Ground Segmentation and Traversability Estimation for Terrestrial Robots,” *Int. J. Control, Automat. Syst. (IJCAS)*, vol. 22, pp. 347–359, Feb. 1, 2024.
9. **Hyungtae Lim**, Beomsoo Kim, Daebeom Kim, Eungchang Mason Lee, and Hyun Myung, “Quatro++: Robust Global Registration Exploiting Ground Segmentation for Loop Closing in LiDAR SLAM,” *Int. J. Robot. Res. (IJRR)*, vol. 43, no. 5, pp. 685–715, Nov. 2, 2023.
10. Alex Junho Lee, Seungwon Song, **Hyungtae Lim**, Wooju Lee, and Hyun Myung, “ $(LC)^2$: LiDAR-Camera Loop Constraints For Cross-Modal Place Recognition,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 8, no. 6, pp. 3589–3596, Apr. 20, 2023.
11. DongKi Noh, Changki Sung, Teayoung Uhm, Wooju Lee, **Hyungtae Lim**, Jaeseok Choi, Kyuewang Lee, Dasol Hong, Daeho Um, Inseop Chung, Hochul Shin, MinJung Kim, Hyoung-Rock Kim, SeungMin Baek, and Hyun Myung, “X-MAS: Extremely Large-Scale Multi-Modal Sensor Dataset for Outdoor Surveillance in Real Environments,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 8, no. 2, pp. 1093–1100, Jan. 12, 2023.
12. Seungwon Song, **Hyungtae Lim**, Alex Junho Lee, and Hyun Myung, “DynaVINS: A Visual-Inertial SLAM for Dynamic Environments,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 4, pp. 11523–11530, Oct. 31, 2022.
13. Minho Oh, Euigon Jung, **Hyungtae Lim**, Wonho Song, Sumin Hu, Eungchang Mason Lee, Junghee Park, Jaekyung Kim, Jangwoo Lee, and Hyun Myung, “TRAVEL: Traversable Ground and Above-Ground Object Segmentation Using Graph Representation of 3D LiDAR Scans,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 3, pp. 7255–7262, Jun. 13, 2022. **Won 2022 IEEE RA-L Best Paper Award.**
14. Seungwon Song, **Hyungtae Lim**, and Hyun Myung, “G2P-SLAM: Generalized RGB-D SLAM Framework for Mobile Robots in Low-Dynamic Environments,” *IEEE Access*, vol. 10, pp. 21370–21383, Feb. 11, 2022.
15. Changki Sung, Seulgi Jeon, **Hyungtae Lim**, and Hyun Myung, “What If There Was No Revisit? Large-Scale Graph-based SLAM with Traffic Sign Detection in an HD Map Using LiDAR Inertial Odometry,” *J. Intell. Serv. Robot.*, pp. 1–10, Nov. 25, 2021.

16. **Hyungtae Lim**, Hanseok Ryu, Matthew Rhudy, Dongjin Lee, Dongjin Jang, Changho Lee, Young-min Park, Wonkeun Youn, and Hyun Myung, “Deep Learning-Aided Synthetic Airspeed Estimation of UAVs for Analytical Redundancy with a Temporal Convolutional Network,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 1, pp. 17–24, Oct. 1, 2021.
17. **Hyungtae Lim**, Minh Oh, and Hyun Myung, “Patchwork: Concentric Zone-Based Region-Wise Ground Segmentation With Ground Likelihood Estimation Using a 3D LiDAR Sensor,” *IEEE Robot. Automat. Lett. (RA-L) with IROS*, vol. 6, no. 4, pp. 6458–6465, Jun. 28, 2021.
18. Wonkeun Youn, **Hyungtae Lim**, Hyoung Sik Choi, Matthew B. Rhudy, Hyeok Ryu, Sungyug Kim, and Hyun Myung, “State Estimation of HALE UAV With Deep Learning-Aided Virtual AOA/SSA Sensor for Analytical Redundancy,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 6, no. 3, pp. 5276–5283, Apr. 19, 2021.
19. **Hyungtae Lim**, Sungwon Hwang, and Hyun Myung, “ERASOR: Egocentric Ratio of Pseudo Occupancy-Based Dynamic Object Removal for Static 3D Point Cloud Map Building,” *IEEE Robot. Automat. Lett. (RA-L) with ICRA*, vol. 6, no. 2, pp. 2272–2279, Feb. 23, 2021.

INTERNATIONAL CONFERENCE

1. Junho Choi, Kihwan Ryoo, Jeewon Kim, Taeyun Kim, Eungchang Lee, Myeongwoo Jeong, Kevin Christiansen Marsim, **Hyungtae Lim**, and Hyun Myung, “SaWa-ML: Structure-Aware Pose Correction and Weight Adaptation-Based Robust Multi-Robot Localization,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, 2025. Accepted. To appear.
2. Dominic Maggio, **Hyungtae Lim**, and Luca Carlone, “VGGT-SLAM: Dense RGB SLAM Optimized on the SL(4) Manifold,” in *Proc. Neural Information Processing Systems (NeurIPS)*, 2025. Accepted. To appear.
3. Minkyun Seo, **Hyungtae Lim**, Kanghee Lee, Luca Carlone, and Jaesik Park, “BUFFER-X: Towards Zero-Shot Point Cloud Registration in Diverse Scenes,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 3851–3862.
4. **Hyungtae Lim**, Daebeom Kim, Gunhee Shin, Jingnan Shi, Ignacio Vizzo, Hyun Myung, Jaesik Park, and Luca Carlone, “KISS-Matcher: Fast and Robust Point Cloud Registration Revisited,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Atlanta, U.S., May 19–23, 2025, pp. 11104–11111.
5. Kihwan Ryoo, **Hyungtae Lim**, and Hyun Myung, “MambaGlue: Fast and Robust Local Feature Matching With Mamba,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Atlanta, U.S., May 19–23, 2025. Accepted. To appear.
6. **Hyungtae Lim**, Seoyeon Jang, Benedikt Mersch, Jens Behley, Hyun Myung, and Cyrill Stachniss, “HeLiMOS: A Dataset for Moving Object Segmentation in 3D Point Clouds From Heterogeneous LiDAR Sensors,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Abu Dhabi, UAE, Oct. 14–18, 2024.
7. Changki Sung, Wanhee Kim, Jungho An, Wooju Lee, **Hyungtae Lim**, and Hyun Myung, “Contextrast: Contextual Contrastive Learning for Semantic Segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, U.S., Jun. 17–21, 2024, pp. 3732–3742.
8. Minh Oh, Gunhee Shin, Seoyeon Jang, Seungjae Lee, Dongkyu Lee, Wonho Song, Byeongho Yu, **Hyungtae Lim**, Jaeyoung Lee, and Hyun Myung, “B-TMS: Bayesian Traversable Terrain Modeling and Segmentation Across 3D LiDAR Scans and Maps for Enhanced Off-Road Navigation,” in *Proc. IEEE Intell. Veh. (IV)*, Jeju Island, Korea, Republic of, Jun. 2–5, 2024.
9. Wooju Lee, Dasol Hong, **Hyungtae Lim**, and Hyun Myung, “Object-Aware Domain Generalization for Object Detection,” in *Annual AAAI Conf. Artif. Intell. (AAAI)*, Vancouver, Canada, Feb. 20–27, 2024, pp. 2946–2955.
10. Seoyeon Jang, Minh Oh, Byeongho Yu, I Made Aswin Nahrendra, Seungjae Lee, **Hyungtae Lim**, and Hyun Myung, “TOSS: Real-time Tracking and Moving Object Segmentation for Static Scene Mapping,” in *Proc. Int. Conf. Robot Intell. Tech. Applications (RiTA)*, Xian, China, Dec. 6–8, 2023. **Received Best Paper Award.**
11. Dong-Uk Seo, **Hyungtae Lim**, Eungchang Mason Lee, Hyunjun Lim, and Hyun Myung, “Enhancing Robustness of Line Tracking Through Semi-Dense Epipolar Search in Line-based SLAM,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Detroit, U.S., Oct. 1–5, 2023, pp. 3483–3490.
12. **Hyungtae Lim**, Lucas Nunes, Benedikt Mersch, Xieyuanli Chen, Jens Behley, Hyun Myung, and Cyrill Stachniss, “ERASOR2: Instance-Aware Robust 3D Mapping of the Static World in Dynamic Scenes,” in *Robotics: Science and Systems (RSS)*, Daegu, South Korea, Jul. 15–19, 2023.

13. **Hyungtae Lim**, Daebeom Kim, Beomsoo Kim, and Hyun Myung, “AdaLIO: Robust Adaptive LiDAR-Inertial Odometry in Degenerate Indoor Environments,” in *Proc. Int. Conf. Ubiquit. Robot. (UR)*, Hawaii, U.S., Jun. 25–28, 2023, pp. 48–53.
14. **Hyungtae Lim**, Kawon Han, Gunhee Shin, Giseop Kim, Songcheol Hong, and Hyun Myung, “ORORA: Outlier-Robust Radar Odometry,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, London, UK, May 29–Jun. 2, 2023, pp. 2046–2053.
15. Alex Junho Lee, **Hyungtae Lim**, Minho Oh, Wonho Song, and Hyun Myung, “Volumetric Vegetation Monitoring From LiDAR Scans With Ground Estimation,” in *Proc. Int. Conf. Control, Automat. Syst. (ICCAS)*, Busan, Republic of Korea, Nov. 27–Dec. 1, 2022, pp. 1378–1379.
16. Seungjae Lee, **Hyungtae Lim**, and Hyun Myung, “Patchwork++: Fast and Robust Ground Segmentation Solving Partial Under-Segmentation Using 3D point cloud,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Kyoto, Japan, Oct. 22–27, 2022, pp. 13276–13283.
17. Sumin Hu, Yeeun Kim, **Hyungtae Lim**, Alex Junho Lee, and Hyun Myung, “eCDT: Event Clustering for Simultaneous Feature Detection and Tracking,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Kyoto, Japan, Oct. 22–27, 2022, pp. 3808–3815.
18. Dong-Uk Seo, **Hyungtae Lim**, Seungjae Lee, and Hyun Myung, “PaGO-LOAM: Robust ground- optimized LiDAR odometry,” in *Proc. Int. Conf. Ubiquit. Robot. (UR)*, Jeju, Republic of Korea, Jul. 4–6, 2022, pp. 1–7.
19. **Hyungtae Lim**, Suyong Yeon, Soohyun Ryu, Yonghan Lee, Youngji Kim, Jaeseong Yun, Euigon Jung, Donghwan Lee, and Hyun Myung, “A Single Correspondence Is Enough: Robust Global Registration to Avoid Degeneracy in Urban Environments,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Philadelphia, USA, May 23–27, 2022, pp. 8010–8017.
20. Sungwon Hwang, **Hyungtae Lim**, and Hyun Myung, “Equivariance-Bridged SO(2)-Invariant Representation Learning Using Graph Convolutional Network,” in *Proc. British Machine Vis. Conf. (BMVC)*, Virtual, Oct. 22–25, 2021.
21. Hyungyu Lee, Myeongwoo Jeong, Chanyoung Kim, **Hyungtae Lim**, Changgwe Park, Sungwon Hwang, and Hyun Myung, “Low-level Pose Control of Tilting Multirotor for Wall Perching Tasks Using Reinforcement Learning,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Prague, Czech Republic (Virtual), Sep. 27–Oct. 1, 2021, pp. 9669–9676.
22. Eungchang Mason Lee, Junho Choi, **Hyungtae Lim**, and Hyun Myung, “REAL: Rapid Exploration With Active Loop-Closing Toward Large-Scale 3D Mapping Using UAVs,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Prague, Czech Republic (Virtual), Sep. 27–Oct. 1, 2021, pp. 4194–4198.
23. **Hyungtae Lim**, Hyeonjae Gil, and Hyun Myung, “MSDPN: Monocular Depth Prediction With Partial Laser Observation Using Multi-stage Neural Networks,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Las Vegas, USA (Virtual), Oct. 24–Dec. 24, 2020, pp. 10750–10757.
24. **Hyungtae Lim**, Sungwon Hwang, Sungjae Shin, and Hyun Myung, “Normal Distributions Transform is Enough: Real-Time 3D Scan Matching for Pose Correction of Mobile Robot Under Large Odometry Uncertainties,” in *Proc. Int. Conf. Control, Automat. Syst. (ICCAS)*, Busan, Republic of Korea, Oct. 13–16, 2020, pp. 1155–1161, **Received Student Best Paper Award**.
25. **Hyungtae Lim**, Changgwe Park, and Hyun Myung, “RONet: Real-Time Range-Only Indoor Localization via Stacked Bidirectional LSTM with Residual Attention,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Macau, China, Nov. 4–8, 2019, pp. 3241–3247.
26. Jungmo Koo, Changgwe Park, **Hyungtae Lim**, and Hyun Myung, “Light-Weight Deep Neural Networks for Multi-Target Classification,” in *Proc. Int. Conf. Control, Automat. Syst. (ICCAS)*, Jeju, Republic of Korea, Oct. 15–18, 2019.
27. Jieum Hyun, Taekjun Oh, **Hyungtae Lim**, and Hyun Myung, “UWB-Based Indoor Localization Using Ray-Tracing Algorithm,” in *Proc. Int. Conf. Ubiquit. Robot. (UR)*, Hawaii, USA, Jun. 24–27, 2019, pp. 98–101.
28. **Hyungtae Lim**, Jungmo Koo, Jieum Hyun, and Hyun Myung, “Effective Indoor Robot Localization by Stacked Bidirectional LSTM Using Beacon-Based Range Measurements,” in *Proc. Int. Conf. Robot Intell. Tech. Applications (RITA)*, Putrajaya, Malaysia, Dec. 16–18, 2018, pp. 144–151.
29. **Hyungtae Lim**, Jungmo Koo, Jieum Hyun, and Hyun Myung, “Recurrent Neural Networks for Range-Based Indoor Robot Localization,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Madrid, Spain, Oct. 1–5, 2018.

Teaching Experience

[CTP445] Augmented Reality (Lecturer: Prof. Woontack Woo) GUEST LECTURER FOR SIX WEEKS TO TEACH BASIC KNOWLEDGE OF VISUAL SLAM	<i>Minor Program in Culture Technology, KAIST</i> Spring, 2023
[EE581] Linear Systems (Lecturer: Prof. Donghwan Lee) TEACHING ASSISTANT	<i>Electrical Engineering, KAIST</i> Spring, 2022
[EE585] Mobile Robotics and Autonomous Navigation (Lecturer: Prof. Hyun Myung) TEACHING ASSISTANT	<i>Electrical Engineering, KAIST</i> Fall, 2021
[EE581] Linear Systems (Lecturer: Prof. Donghwan Lee) TEACHING ASSISTANT	<i>Electrical Engineering, KAIST</i> Spring, 2021
[EE688] Optimal Control Theory (Lecturer: Prof. Donghwan Lee) TEACHING ASSISTANT	<i>Electrical Engineering, KAIST</i> Fall, 2020
[EE305] Introduction to Electronics Design Lab (Lecturer: Prof. Dongho Cho) TEACHING ASSISTANT	<i>Electrical Engineering, KAIST</i> Spring, 2020
[CE208] IT in Construction Engineering (Lecturer: Prof. Hyun Myung) TEACHING ASSISTANT	<i>Civil Engineering, KAIST</i> Fall, 2018
[CE558] Introduction to Civil Robotics (Lecturer: Prof. Hyun Myung) TEACHING ASSISTANT	<i>Civil Engineering, KAIST</i> Spring, 2018

Honors & Awards

INTERNATIONAL AWARDS

2025	IEEE ICRA'25 Outstanding Reviewer Award , IEEE ICRA'25	<i>Atlanta, USA</i>
2024	RSS Pioneer , RSS Pioneers 2024 Workshop, in RSS'24	<i>Delft, Netherlands</i>
2024	1st Prize , HILTI SLAM Challenge in <i>IEEE ICRA'24</i>	<i>Yokohama, Japan</i>
2023	Best Paper Award , RiTA'23	<i>Xian, China</i>
2023	Best Paper Award (among 1,100 papers, 5 papers are received) , IEEE RA-L	<i>London, UK</i>
2023	1st Prize (among 63 teams) , HILTI SLAM Challenge in <i>IEEE ICRA'23</i>	<i>London, UK</i>
2023	CES Innovation Award , Consumer Technology Association (CES)	<i>San Francisco, USA</i>
2022	2nd Cash Prize, 4th in Total , HILTI SLAM Challenge in <i>IEEE ICRA'22</i>	<i>Philadelphia, USA</i>
2020	Student Best Paper Award , ICCAS	<i>Busan, Republic of Korea</i>

DOMESTIC AWARDS

2020	Kim Sung-Bue Creative Activity Award , KAIST	<i>Daejeon</i>
2019	Han Cheolhui Augustine Scholarship , EE, KAIST	<i>Daejeon</i>
2019	1st Prize , Hitachi-LG LiDAR Application Competition	<i>Seoul</i>
2018	Excellence Prize , Smart City Service and Start-Up Competition	<i>Seoul</i>
2015	Dean's List , ME, KAIST	<i>Daejeon</i>

Academic Services

2025	Program committee , RSS Pioneer'2025 Workshop committee	<i>USA</i>
2024–	Associate Editor , IEEE Robot. Automat. Lett. (RA-L)	
2025	Reviewer , RA-L, T-RO, IJRR, IROS, ICRA, CVPR, NeurIPS	
2024	Reviewer , RA-L, T-RO, IJRR, IROS, ICRA, CVPR	
2023	Reviewer , RA-L, T-RO, IJRR, IROS, ICRA	